

How to use this book

Standing methodological rules

Three rules apply throughout and are not restated in each chapter.

1. **Every stochastic result is reported across at least 20 seeds.** t-SNE and UMAP are seed-dependent; a single run is an anecdote. Seeds are fixed in `_common.R` as `BENCH_SEEDS` and disclosed with every figure.
2. **Every θ -dependent result is reported as a curve, never as a single number.** Fold angle is a continuous difficulty parameter. Reporting one value invites cherry-picking, and hides exactly the regime changes the benchmark exists to expose.
3. **Every figure carries `fig.alt`.** No result is encoded by colour alone.

What you need

R 4.1 or newer and the packages pinned in `renv.lock`. Run `renv::restore()` once after cloning. The companion package `foldbench` provides the pattern constructors, the sampler, and the evaluation metrics; the book loads it but does not reproduce its source.

The Chapter @ref(results) benchmark grid takes hours to compute. It is precomputed and committed under `data/processed/`; `scripts/run-benchmark-grid.R` is kept as the provenance record rather than run during the build.

How to read it

Part I builds the object. Part II tests four families of method against it. Part III is the benchmark proper and contains the book's main contribution — Chapter @ref(evaluation), where the evaluation metrics themselves are checked against exact truth. Part IV asks whether any of it changes a real decision.

Readers who only want the benchmark can start at Chapter @ref(benchmarks) and refer back to Chapter @ref(generative) for the generative model.